

CONSTRAINT-FIRST TOPOLOGY:
Foam Cobordisms, Nilpotent Transport,
and Immersed Singularities as Admissibility Geometry

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Abstract

Three recent papers in low-dimensional topology and homological algebra — the foam cobordism theory of Im and Khovanov, the nilpotent pair counting of Chen, Im, Khovanov, Lillja, and Rugo, and the immersed surface functoriality of Carter, Cooper, Khovanov, and Krushkal — independently rediscover a recurring invariant architecture that we term *constraint-first process geometry*. In each domain, meaningful structure does not reside in static objects but emerges as equivalence classes of admissible local transformations under globally consistent constraint systems. Weighted foam cobordisms encode conserved interval rearrangements whose invariant lives in the exterior algebra $\mathbb{R} \wedge_{\mathbb{Q}} \mathbb{R}$; nilpotent operator pairs classify constrained bidirectional transport by depth-symmetric propagation modes whose extinction behavior mirrors finite-time entropy collapse; and immersed surface cobordisms with transverse double points extend categorical functoriality by admitting controlled singularities as structured morphisms rather than pathological failures of embedding.

We interpret all three constructions through the lens of the Relativistic Scalar-Vector Plenum (RSVP) framework and the Constraint-Leveraged Inference Operator (CLIO), arguing that these mathematical systems independently formalize three operational faces of a single underlying architecture: process-primacy, obstruction-sensitivity, and admissibility-governed invariant extraction. CLIO is introduced informally through its operational role in each domain before being crystallized as a formal operator in the synthesis. Our central claim is that constraint-first systems do not derive coherence from objects; rather, objects emerge as stable equivalence classes of admissible transformations.

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1 Introduction: The Recurring Architecture

There is a structural pattern appearing independently across several branches of contemporary mathematics that is not yet widely recognized as a unified phenomenon. The pattern concerns systems in which the fundamental entities are not objects but transformations, and in which meaning is carried not by intrinsic properties of states but by invariants preserved under classes of admissible local operations. Such systems share three distinguishing features. First, they are *process-primary*: the ontologically basic entity is a transformation history or equivalence class thereof, not an instantaneous configuration. Second, they are *obstruction-sensitive*: the invariants that survive constraint-preserving operations are precisely those that encode what cannot be eliminated by any sequence of locally admissible moves. Third, they are *admissibility-governed*: coherence of the global structure is guaranteed not by metric or embedding data but by the consistent closure of local rewrite or compatibility laws.

Three papers in low-dimensional topology and linear algebra serve as our primary mathematical sources. The foam cobordism paper of Im and Khovanov [1] develops a theory in which weighted oriented one-foams — finite combinatorial graphs whose edges carry positive real labels with conservation-compatible merge and split rules at vertices — are organized into cobordism equivalence classes. Their central result identifies the cobordism group of such foams with the exterior product $\mathbb{R} \wedge_{\mathbb{Q}} \mathbb{R}$, the abelianization of the group of interval exchange transformations via the Sah–Arnoux–Fathi (SAF) invariant, and the first K-theory group of the Zakharevich assembler category. The identification is not merely algebraic: it reveals that the combinatorial topology of weighted seams and cobordism moves is a faithful encoding of interval exchange dynamics and its associated irreversibility obstruction.

The nilpotent pair paper of Chen, Im, Khovanov, Lillja, and Rugo [2] extends a beautiful observation of Leinster concerning nilpotent operators on a single vector space to pairs of linear maps between two spaces linked by bidirectional composition. A pair $(f : V \rightarrow W, g : W \rightarrow V)$ is called nilpotent when the compositions gf and fg are both nilpotent. The paper introduces a key conceptual innovation — the distinction between *balanced* and *unbalanced* vectors — which classifies indecomposable nilpotent representations of the oriented two-cycle quiver. This classification yields a closed counting formula over finite fields and a strikingly elegant set-theoretic analogue connecting eventual constancy of map pairs with spanning trees in complete bipartite graphs.

The immersed cobordism paper of Carter, Cooper, Khovanov, and Krushkal

[3] extends Khovanov homology from smooth embedded surface cobordisms in four-space to immersed surfaces possessing transverse double-point singularities. Rather than treating double points as failures of embedding, the paper assigns specific chain maps A and B to positive and negative nodes, extends the Carter–Saito movie move calculus to include node interactions, and proves that the resulting 2-functor remains well-defined on isotopy classes of immersed surfaces. The homological content of the singular transitions is directly encoded by the Khovanov homology of the Hopf link.

Our purpose in this essay is not to compress these papers into a single formal system but to exhibit a structural resonance among them that points toward a common underlying architecture. We interpret this architecture through two complementary frameworks developed in prior work: the *Relativistic Scalar-Vector Plenum* (RSVP), a field-theoretic model of admissibility-governed process evolution, and the *Constraint-Leveraged Inference Operator* (CLIO), an abstract operator class governing constrained projection and trajectory compression. Both frameworks share the central conviction that coherent structure — physical, semantic, or topological — is not a property of configurations but of admissible transformations between them.

The essay proceeds in four parts. Part I examines foam cobordisms as constraint-closure invariants and the SAF obstruction as a topological irreversibility detector. Part II examines nilpotent pair transport as finite-time entropy exhaustion and balanced vectors as coherence-preserving propagation modes. Part III examines immersed surface cobordisms as an extension of categorical morphism space admitting controlled singular transitions. Part IV synthesizes these correspondences and presents CLIO in its formal role, arguing that all three mathematical systems independently instantiate the same constraint-first process geometry.

Throughout, we distinguish three layers of structure that appear in each domain: local transition laws governing individual moves, constraint-preserving global invariants encoding obstruction information, and admissibility geometry determining which local transformations preserve coherent global structure. This tripartite organization is not imposed but discovered, appearing in each of the three source papers nearly uncannily well.

2 Part I: Foam Cobordisms as Constraint-Closure Invariants

2.1 Weighted foams and conserved interval decomposition

A weighted oriented one-foam in the sense of Im and Khovanov is a finite oriented trivalent graph whose edges carry positive real labels, subject to the conservation condition that at every merge or split vertex the labels sum consistently: if two edges of weights a and b merge into a single edge, that edge carries weight $a + b$. This is not a metric constraint. The number a does not measure the geometric length of an edge in any embedding space. It is a *compatibility label* ensuring that the vertex interaction respects a local conservation law. The paper is explicit about this: the thickening constructions in Sections 2.4.1 and 2.4.2 of [1] interpret these labels as measures of interval width in the lower-limit topology or as weights in a measured foliation, but the algebraic structure is entirely independent of any embedding.

This is the first appearance of what we will call *constraint-before-content*: the label a is meaningful not as an intrinsic property of an edge in isolation but only relative to the compatibility constraints imposed at vertices. An edge label acquires significance through its participation in admissible merge and split events.

Formally, given a commutative semigroup $(H, +)$, one defines the cobordism group Cob_H^1 of H -weighted oriented one-foams modulo H -weighted oriented two-foam cobordisms. The main theorem of Section 2 of [1] then asserts:

Theorem 2.1 (Im–Khovanov [1]). *There are isomorphisms of abelian groups*

$$\text{Cob}_{>0}^1 \cong \mathbb{R} \wedge_{\mathbb{Q}} \mathbb{R} \cong H_1(\text{Aut}_{\text{IET}}, \mathbb{Z}) \cong K_1(\mathcal{C}_Z),$$

where $\text{Cob}_{>0}^1$ denotes the cobordism group of weighted oriented one-foams with positive real labels, Aut_{IET} is the group of interval exchange transformations of $[0, 1)$, $\mathcal{C}_Z$ is the Zakharevich assembler category for interval exchange transformations, and the second isomorphism is the Sah–Arnoux–Fathi invariant.

The proof proceeds by constructing an isomorphism $\psi : \text{Cob}_{>0}^1 \xrightarrow{\sim} \mathbb{R} \wedge_{\mathbb{Q}} \mathbb{R}$ via a local invariant ν assigned to planar projections of one-foams. The invariant accumulates contributions from merge and split vertices according to their thin-edge orderings and from virtual crossings between interval strands. One key calculation is that the invariance of ν under move 1 in Figure 15 of [1] requires exactly the relation $a \wedge b + b \wedge a = 0$, and invariance under move 5 requires the bilinearity of the tensor product. The cobordism relations are thus the geometric

shadows of the algebraic relations defining $\mathbb{R} \wedge_{\mathbb{Q}} \mathbb{R}$.

2.2 The SAF invariant as topological irreversibility

The Sah–Arnoux–Fathi invariant of an interval exchange transformation $T_{\lambda, \sigma}$ is defined as

$$\text{SAF}(T_{\lambda, \sigma}) = \sum_{i=1}^r \lambda_i \otimes t_i = 2 \sum_{i < j : \sigma(j) < \sigma(i)} \lambda_i \wedge \lambda_j,$$

where t_i is the displacement of the i -th interval under σ . It vanishes precisely on the commutator subgroup of Aut_{IET} , yielding the short exact sequence

$$1 \rightarrow [\text{Aut}_{\text{IET}}, \text{Aut}_{\text{IET}}] \rightarrow \text{Aut}_{\text{IET}} \xrightarrow{\text{SAF}} \mathbb{R} \wedge_{\mathbb{Q}} \mathbb{R} \rightarrow 1.$$

The invariant detects the *irremovable asymmetry* of an interval permutation: those rearrangements that cannot be expressed as products of commutators are precisely those whose SAF class is nontrivial in the exterior algebra.

From the perspective of RSVP admissibility geometry, this is structurally very close to a topological entropy circulation invariant. The system locally permutes intervals without altering total measure, yet globally there persists a conserved antisymmetric quantity that survives all locally admissible rearrangements. In RSVP, similarly, field configurations related by admissible entropy-compatible evolution share the same effective obstruction class, and what distinguishes genuine physical trajectories from arbitrary rearrangements of configuration data is the preservation of such circulation invariants.

CORRESPONDENCE PRINCIPLE 2.2 (SAF Obstruction and Constraint-Closure Geometry). *The Sah–Arnoux–Fathi invariant of an interval exchange transformation is the minimal admissibility invariant of interval rearrangement dynamics: it encodes exactly that asymmetric relational structure which no sequence of locally admissible reassociations can eliminate. In RSVP-style admissibility geometry, this corresponds to the role of topological obstruction classes in trajectory compression: the SAF class is the residual entropy-circulation datum surviving constraint-preserving coarse-graining.*

The wedge structure $\lambda_i \wedge \lambda_j$ is especially significant. It is antisymmetric in the pair (λ_i, λ_j) , so it records the *order* of the two intervals relative to the permutation, not merely their coexistence. This is not metric information. It is *relational orientation information* surviving the abelianization: the class $\lambda_i \wedge \lambda_j \in \mathbb{R} \wedge_{\mathbb{Q}} \mathbb{R}$ witnesses the fact that i and j were inverted by σ in a way that no local cobordism can undo.

2.3 The antisymmetric bracket and local reassociation laws

Section 3 of [1] develops a parallel theory for planar unoriented weighted one-foams. The cobordism group $\text{Cob}_{>0}^{1,\text{up}}$ of such foams is generated by tripod foams $T(a, b)$ and is isomorphic to the group $Z^2(\mathbb{R}_{>0})$ defined by generators $[a, b]$ with $a, b > 0$ and the relations

$$[a, a] = 0, \tag{1}$$

$$[a, b] + [b, a] = 0, \tag{2}$$

$$[a, b] + [a + b, c] = [a, b + c] + [b, c]. \tag{3}$$

Relation (3) is the defining identity of this algebraic structure. Im and Khovanov observe that it is a two-cocycle relation: it can be interpreted as the signed boundary of a three-simplex with oriented edges labeled by $a, b, c, a + b, b + c, a + b + c$. Geometrically, it expresses the commutativity of two distinct sequences of vertex cobordisms merging three parallel strands of weights a, b, c into a single strand of weight $a + b + c$.

From our perspective, relation (3) is a *local reassociation law*. It asserts that two distinct orderings of a three-way merge are equivalent in the cobordism group. This is formally very close to the Mac Lane pentagon coherence condition in monoidal category theory, and it resembles the entropy redistribution identities appearing in TARTAN recursive tiling: the decomposition order changes, but the global constraint-compatible invariant does not.

STRUCTURAL EQUIVALENCE PRINCIPLE 2.3 (Reassociation Invariance and TARTAN Constraint Closure). *The antisymmetric two-cocycle relation (3) is the foam-theoretic instance of a general reassociation invariance principle: any two admissible sequences of local decomposition events producing the same boundary configuration are equivalent under constraint-consistent cobordism. In the TARTAN framework, an analogous principle governs recursive tiling and constraint propagation: distinct local rewrite sequences preserving boundary admissibility conditions are identified in the global constraint-closure quotient.*

The bracket $[a, b]$ is *almost* bilinear: Proposition 3.5 of [1] shows that $2([a, b_1 + b_2] - [a, b_1] - [a, b_2]) = 0$ in $Z^2(\mathbb{R}_{>0})$. The deviation from linearity is a two-torsion obstruction. In RSVP terms, this two-torsion corresponds to a residual orientational ambiguity in the constraint-closure quotient: the system is almost, but not quite, indifferent to linear decomposition order.

2.4 Cobordism classes as admissibility equivalences

The most conceptually important move in [1] is the replacement of geometric identity with cobordism equivalence. Two weighted one-foams are declared equivalent not when they are homeomorphic as graphs with labels but when one can be transformed into the other through a sequence of elementary two-foam cobordisms: cap and cup moves adding or removing circles, saddle cobordisms, vertex cobordisms passing through a seam interaction, and isotopies of planar diagrams. Each elementary cobordism preserves the local conservation law at seams. The global equivalence class encodes exactly what cannot be undone by any sequence of such moves.

This is the foam-theoretic realization of what we mean by a *constraint-first ontology*. The meaningful object is not the weighted graph itself but its equivalence class under admissible local operations. The operations are primary; the equivalence class — the cobordism group element — is derived. Obstruction to triviality is encoded by the SAF invariant or the bracket class $[a, b]$, and these are precisely the relational data that survive all locally admissible transformations.

In the lower-limit-topology thickening of Section 2.4.1 of [1], a weighted one-foam U is replaced by a topological space $T(U)$ locally homeomorphic to $[0, 1)_\ell \times (0, 1)$, where $[0, 1)_\ell$ carries the lower-limit topology. The foliation of $T(U)$ by leaves $x \times (0, 1)$ gives a measured foliation in which the leaf-compactness condition (all leaves being closed circles) is exactly the condition for U to be null-cobordant. This is a local-to-global theorem mediated by admissibility: what is geometrically local (leaf compactness) determines what is globally trivial (cobordism class).

3 Part II: Nilpotent Transport and Finite-Time Entropy Exhaustion

3.1 Bidirectional transport and quiver representations

The paper [2] studies pairs of linear maps $(f : V \rightarrow W, g : W \rightarrow V)$ between finite-dimensional vector spaces, asking when the composed endomorphisms $T = gf : V \rightarrow V$ and $T' = fg : W \rightarrow W$ are both nilpotent. This is a representation-theoretic question: such a pair is precisely a representation of the oriented two-cycle quiver

$$\Gamma = (V \underset{g}{\overset{f}{\rightrightarrows}} W),$$

and nilpotency of the compositions is the nilpotency condition on the representation.

The shift from a single endomorphism to a pair of maps in opposite directions is mathematically significant. For a single nilpotent endomorphism $T : V \rightarrow V$, the indecomposable representations are Jordan blocks J_r of varying sizes $r \geq 1$, each characterized by a single integer. For the two-cycle quiver, indecomposable nilpotent representations are classified by a pair of integers (m, n) with $|m - n| \leq 1$, reflecting the bidirectional transport structure.

From the RSVP perspective, this shift models the transition from a single state manifold with self-referential evolution to a coupled pair of manifolds linked by transport operators. The compositions $T = gf$ and $T' = fg$ are not the transports themselves but the *round-trip operators* measuring what survives a complete transit in both directions. Nilpotency of T means that iterated round-trips eventually annihilate all state vectors: the reachable future volume under repeated bidirectional transport shrinks to zero in finite time.

3.2 Balanced and unbalanced vectors as propagation modes

The central conceptual innovation of [2] is the distinction between balanced and unbalanced vectors. Given a nilpotent pair (f, g) and a vector $v \in V$, one forms two ascending chains of iterates:

$$v, Tv, T^2v, \dots, T^{a-1}v \neq 0, \quad T^a v = 0$$

$$fv, T'fv, (T')^2fv, \dots, (T')^{\ell-1}fv \neq 0, \quad (T')^\ell fv = 0.$$

The vector v is *balanced* if $a = \ell$, meaning the two chains have equal length and f restricts to an isomorphism $T[v] \xrightarrow{\sim} T'[fv]$ between the corresponding cyclic subspaces. It is *unbalanced* if $\ell = a - 1$, meaning the image chain is shorter by exactly one step.

This dichotomy classifies the propagation modes of the bidirectional transport system. A balanced vector is a mode in which information propagates symmetrically through the coupled system: the depth of recursive iteration is equal in both spaces, and the transport map f identifies the two recursive histories. An unbalanced vector is a dissipative mode: information propagates one step further in the source space V than in the target space W , producing an asymmetric accessibility profile.

CORRESPONDENCE PRINCIPLE 3.1 (Balanced Propagation and Coherent Transport).

Balanced vectors in a nilpotent pair (f, g) are the constraint-preserving propagation modes of bidirectional transport: they witness symmetric recursive iteration depth across the coupled manifolds V and W and correspond to coherent transport channels in which no dimensional collapse asymmetry arises. Unbalanced vectors are dissipative modes encoding one-sided information loss. In RSVP admissibility geometry, this distinction corresponds to the decomposition of transport operators into coherence-preserving and entropy-producing sectors.

The paper's generalization of Leinster's bijection is the following result.

Theorem 3.2 (Chen–Im–Khovanov–Lillja–Rugo [2]). *There is a bijection*

$$\mathrm{Hom}(V, W) \times \mathrm{Hom}(W, V) \times (V \cup_0 W) \cong \mathcal{N}(V, W) \times V \times W,$$

where $\mathcal{N}(V, W)$ denotes the set of nilpotent pairs and $V \cup_0 W$ is the union of V and W along their zero vectors. Over a finite field $\mathbb{F}_q$, this yields the closed formula

$$N_{m,n} = q^{2mn-m-n}(q^m + q^n - 1)$$

for the number of nilpotent pairs with $\dim V = m$ and $\dim W = n$.

The proof passes through Lemma 3.14 of [2], which constructs an explicit bijection between quadruples with balanced components and quadruples with unbalanced components by a controlled redefinition of f on the subspace $T[v_b]$. This bijection is the algebraic heart of the paper and is structurally reminiscent of a gauge transformation: the pair (f, g) is modified on a single subspace to exchange the balanced and unbalanced sectors without changing the global nilpotency class.

3.3 Nilpotency as finite-time entropy exhaustion

The nilpotent condition $T^k = 0$ for some finite k has a clear geometric interpretation in RSVP language. Define the *accessibility volume* of the transport system at depth n as the dimension of $\mathrm{im}(T^n)$. Nilpotency asserts that this volume decreases strictly and reaches zero at the finite depth k . The sequence

$$\dim V = \dim \mathrm{im}(T^0) \geq \dim \mathrm{im}(T^1) \geq \cdots \geq \dim \mathrm{im}(T^k) = 0$$

is an exact analogue of entropy descent: the logarithm of the accessibility volume decreases monotonically to $-\infty$ under iterated application of T .

In RSVP, entropy is defined geometrically as the logarithmic accessible future volume in trajectory space under admissibility constraints. A nilpotent transport

operator is therefore a *finite-time accessibility extinguisher*: a morphism of the state manifold under which all admissible future structure is annihilated within k steps.

STRUCTURAL EQUIVALENCE PRINCIPLE 3.3 (Nilpotent Exhaustion and RSVP Entropy Descent). *A nilpotent pair (f, g) with nilpotency degree k is the algebraic model of a finite-time entropy exhaustion process: the accessibility volume of the bidirectional transport system descends strictly to zero in exactly k iterations of the round-trip operator. This is structurally equivalent to a bounded-depth admissibility collapse in RSVP trajectory geometry, in which the admissible future cone shrinks to a point in finite proper time under constraint-preserving evolution.*

3.4 The set-theoretic model and acyclic causal structure

The set-theoretic analogue developed in Section 2 of [2] is remarkably clean. For finite sets X and Y with $|X| = m$ and $|Y| = n$, a pair of maps $(f : X \rightarrow Y, g : Y \rightarrow X)$ is *eventually constant* when the compositions $(gf)^k$ and $(fg)^k$ eventually map all elements to a unique 2-cycle attractor (x_0, y_0) with $f(x_0) = y_0$ and $g(y_0) = x_0$.

Theorem 2.4 of [2] establishes a bijection between eventually constant pairs and spanning trees of the complete bipartite graph $K(m, n)$ together with a choice of distinguished edge. The counting formula is $m^{n-1}n^{m-1}(m + n - 1)$.

The spanning-tree correspondence is conceptually central. A spanning tree on a bipartite graph is a minimal connected acyclic structure: it contains all vertices, preserves global reachability, and eliminates all cycles. In RSVP and CLIO terms, trees are *minimal admissibility skeletons*: they retain just enough connectivity to support constrained propagation while removing all redundant entropy-producing feedback.

The eventually constant condition is precisely acyclicity of the causal structure outside the minimal 2-cycle attractor. All state transitions flow toward the attractor along tree-like dependency paths. This mirrors the TARTAN principle that stable recursive systems enforce directed acyclic propagation and eliminate self-referential closure loops.

3.5 Boolean nilpotency and directed acyclic causal graphs

The Boolean semiring $\mathbb{B} = \{0, 1 : 1 + 1 = 1\}$ provides the sharpest version of the acyclicity correspondence. Over $\mathbb{B}$, Proposition 4.2 of [2] shows that a Boolean matrix $A \in \text{Mat}_n(\mathbb{B})$ is nilpotent if and only if the directed graph with adjacency matrix A is a directed acyclic graph (DAG). The proof is immediate: a 1 on the diagonal of any power A^k is the algebraic signature of an oriented cycle, and

nilpotency is exactly the condition excluding all such cycles.

INTERPRETIVE PROPOSITION 3.4 (Boolean Nilpotency and Causal Admissibility). *Nilpotent Boolean matrices are in bijection with directed acyclic graphs on n ordered vertices. The nilpotency condition is therefore equivalent to the absence of any self-reinforcing causal cycle in the associated computation graph. In the context of CLIO and RSVP, this formalizes the principle that constraint-admissible recursive computation is acyclic: admissible inference propagation proceeds without pathological self-reference, and entropy-respecting modular composition excludes runaway feedback loops.*

The recurrence $a_n = \sum_{k=1}^n (-1)^{k-1} \binom{n}{k} 2^{k(n-k)} a_{n-k}$ counting DAGs on n ordered vertices (Corollary 4.3 of [2]) encodes the combinatorial complexity of admissible causal structures. Each term in the inclusion-exclusion alternates signs in the manner of an obstruction-theoretic count: positive contributions from acyclic structures, negative corrections removing configurations containing at least one cycle.

4 Part III: Immersed Cobordisms as Controlled Singular Transitions

4.1 Extending the morphism category

The paper [3] addresses a fundamental question in categorical topology: how far can functoriality be extended? Classical Khovanov homology associates to each oriented link $L \subset \mathbb{R}^3$ a bigraded abelian group $\text{Kh}(L)$ and to each smooth oriented surface cobordism $\Sigma \subset \mathbb{R}^3 \times [0, 1]$ between links a graded map $\text{Kh}(L_0) \rightarrow \text{Kh}(L_1)$, well-defined up to overall sign on isotopy classes of surfaces. The 2-functor

$$\kappa : \text{Tang}_b^4 \rightarrow K_p$$

sends balanced tangle cobordisms to chain homotopy classes of maps between Khovanov chain complexes.

The extension problem is: what happens when the surface Σ is no longer smoothly embedded but merely immersed, possessing transverse self-intersection points modeled locally by the complex variety $\{(w, z) \in \mathbb{C}^2 : wz = 0\}$? This local model is the intersection of two complex planes in $\mathbb{C}^2$, whose link in a small 3-sphere is a Hopf link.

Rather than declaring such immersions inadmissible, Carter, Cooper, Khovanov, and Krushkal define chain maps A and B associated to positive and negative

double-point singularities respectively and prove that the resulting assignment extends κ to a 2-functor

$$\tilde{\kappa} : \text{Tang}_{\mathcal{B}}^4 \rightarrow K_p$$

on the larger category $\text{Tang}_{\mathcal{B}}^4$ of balanced immersed cobordisms with double points.

This is a categorical expansion of admissible morphisms: the paper does not embed singular surfaces into a pre-existing smooth theory but genuinely enlarges the 2-category of cobordisms by adding morphism types that carry new algebraic information.

4.2 Singularities as structured computational events

The maps $A : C^+ \rightarrow C^-$ and $B : C^- \rightarrow C^+$ assigned to double points are explicitly computed from the chain complexes associated to positive and negative crossings. The map A has homological degree -2 (in the t -grading) and lives in the class of (t, q) -degree $(-2, -6)$ corresponding to a generator of the q -shifted negative Hopf link homology $q^{-2}H^-$. The map B has degree $+2$ and corresponds to a generator of $q^{-2}H^+$.

The appearance of Hopf link homology here is structurally significant. The Hopf link is the link of the double-point singularity: it bounds a small 3-sphere around the node in 4-space, and its Khovanov homology records the algebraic topology of that local singular region. The maps A and B are therefore not arbitrary choices but are determined by the intrinsic algebraic topology of the singularity type.

INTERPRETIVE PROPOSITION 4.1 (Double Points as Singular Morphism Types). *Transverse double-point singularities in immersed surface cobordisms are not pathological failures of the theory but structured morphism types carrying definite homological information: the chain maps A and B encode the Khovanov homology of the Hopf link, which is the canonical link of the double-point singularity. In RSVP admissibility geometry, this corresponds to the role of entropy concentration loci as structured transition operators rather than excluded defects: a singularity is an admissible event carrying definite obstruction data.*

The homological degree shift is important. Ordinary embedded cobordisms produce degree-zero maps between Khovanov homology groups (after appropriate grading shifts). The singular maps A and B shift the t -degree by ± 2 , allowing access to homological classes that are not reachable by smooth embedded cobordisms. The paper explicitly frames this as a probe of “non-geometric” homology: the

classes in $\text{Kh}^{i,j}(L)$ that cannot be represented as images of smooth surface maps but can be reached through controlled singular transitions.

In RSVP language, this corresponds to the distinction between low-entropy embedded trajectories (which access only a narrow sector of the accessible state space) and controlled entropy-transition operators (which allow passage between sectors separated by admissibility barriers).

4.3 Movie moves as rewrite grammar over admissible transitions

The Carter–Saito movie move framework, extended in Section 4 of [3], is a formal grammar for surface isotopies in four-dimensional space. A movie of a surface $F \subset \mathbb{R}^3 \times [0, 1]$ is a sequence of cross-sectional stills — oriented links in $\mathbb{R}^3 \times \{s_i\}$ — between which a finite list of elementary events occurs: births and deaths of circles, saddle moves, Reidemeister moves of types I, II, and III, and, in the extended theory, double-point creation or annihilation events (nodes).

The movie move theorem (Theorem 4.1 of [3]) asserts that any two movies presenting isotopic immersed surfaces are related by a finite sequence of movie moves from the extended list MM1–MM19. This is a *completeness theorem for the rewrite grammar*: any two equivalent representations of the same isotopy class are connected by finitely many elementary substitutions.

From the perspective of TARTAN and RSVP, this is the topological realization of what we call *trajectory-space equivalence*: the surface is not a static object but an equivalence class of local transition histories, and the movie moves are the rewrite rules identifying different local presentations of the same higher-dimensional process. Different movies are different *semantic foliations* of the same underlying admissibility structure.

STRUCTURAL EQUIVALENCE PRINCIPLE 4.2 (Movie Moves as Gauge Invariance over Presentation). *The extended Carter–Saito movie moves function as gauge equivalences over local presentations of immersed surface cobordisms: any two movies presenting the same isotopy class of immersed surface are related by finitely many elementary rewrite substitutions. In the RSVP and TARTAN frameworks, this is the categorical instance of a general invariance principle: the meaningful entity is not the local presentation (the movie) but the equivalence class of presentations under admissible rewrite operations, and coherence of the global structure is guaranteed by the closure of the rewrite algebra.*

4.4 The chart calculus and projection geometry

The chart formalism developed in Section 4.2 of [3] provides a two-dimensional encoded representation of the full four-dimensional immersed surface. A chart is an oriented labeled graph in the (s, y) -plane encoding, via decorated edges and vertices, the fold singularities, crossing arcs, cusps, triple points, and node events of the surface projected into three-dimensional space.

Formally, the chart is a structured compression of the four-dimensional process into a two-dimensional combinatorial object retaining all information necessary to reconstruct admissible transformations. This is a direct analogue of the CLIO projection

$$\pi : \mathcal{X} \rightarrow \mathcal{M}$$

where $\mathcal{X}$ is the full trajectory space of the surface in $\mathbb{R}^3 \times [0, 1]$ and $\mathcal{M}$ is the compressed representation space carrying the invariant algebraic data. The chart does not record the surface's geometry; it records the admissibility structure of its transformations.

The bookmark code — the combinatorial encoding of a cross-sectional still via symbols for crossings, maxima, minima, and vertical strands — functions as a local coordinate system in $\mathcal{M}$. Different bookmark words for the same link diagram are related by Reidemeister moves, just as different local coordinates on $\mathcal{M}$ are related by the gauge equivalences of the theory.

4.5 Reaching non-geometric homology through singular transitions

A recurrent theme in the topology of four-manifolds is that smooth embedding severely restricts the algebraic invariants accessible to geometric probes. The Whitney trick fails in dimension four: two embedded surfaces in a 4-manifold with algebraically canceling intersection points cannot in general be isotoped apart, and the obstruction is not purely algebraic but depends on the smooth structure. Double points that cannot be removed by homotopy are a fundamental feature of four-dimensional topology rather than an exceptional pathology.

Carter, Cooper, Khovanov, and Krushkal observe that ordinary embedded cobordisms produce maps concentrated near the diagonal of Khovanov homology (along the line $q = 2t + c$ for fixed constants c depending on grading conventions), whereas the singular maps A and B have degrees $(-2, -6)$ and $(2, 4)$ respectively, accessing off-diagonal regions not reachable by smooth cobordisms. Remark 3.4 of [3] frames this as the central motivation: the extension $\text{Tang}_b^4 \subset \text{Tang}_{\hat{b}}^4$ can be

understood as the simplest nontrivial enlargement of the theory along homology classes not represented geometrically.

In RSVP terms: smooth embedded trajectories are *low-entropy morphisms* accessing only a narrow sector of the reachable state space. Controlled singular transitions are *entropy-transition morphisms* crossing admissibility barriers that smooth embeddings cannot traverse. The full accessibility map of Khovanov homology is larger than what smooth geometry alone can explore.

5 Part IV: Synthesis and the CLIO Operator

5.1 The tripartite architecture across three domains

We are now in a position to identify the common architecture appearing in all three source papers. In each domain, there is a precise tripartite organization of the mathematical structure into local transition laws, global constraint invariants, and admissibility geometry. We exhibit this organization explicitly.

In foam cobordism theory, the *local transition laws* are the elementary cobordisms of Figure 16 in [1]: cap and cup moves, saddle cobordisms, vertex cobordisms crossing a seam interaction, and isotopies of planar foam diagrams. These are the primitive admissible local events. The *global constraint invariants* are the SAF class $\text{SAF}(T) \in \mathbb{R} \wedge_{\mathbb{Q}} \mathbb{R}$ and the antisymmetric bracket class $[a, b] \in Z^2(\mathbb{R}_{>0})$, both preserved under all sequences of elementary cobordisms. The *admissibility geometry* is the weighted conservation law at foam vertices: $a + b \leftrightarrow (a, b)$ is the local compatibility constraint determining which elementary cobordisms are permitted.

In nilpotent transport theory, the *local transition laws* are the block-matrix redefinitions of f on cyclic subspaces (as in Lemma 3.14 of [2]), the Fitting decomposition into stable and nilpotent sectors (Lemma 3.3), and the bijective rebalancing between balanced and unbalanced vector sectors. The *global constraint invariants* are the nilpotency degree k of the round-trip operators $T = gf$ and $T' = fg$, the dimension pair $(\dim V_I, \dim V_N)$ of the Fitting decomposition, and the counting invariant $N_{m,n} = q^{2mn-m-n}(q^m + q^n - 1)$. The *admissibility geometry* is the nilpotency condition itself: $T^k = 0$ is the constraint filtering which pairs (f, g) are admissible nilpotent transport systems.

In immersed cobordism theory, the *local transition laws* are the elementary movie events: births, deaths, saddles, Reidemeister moves of types I–III, ψ -moves, and the new double-point node events encoded by A and B . The *global constraint invariants* are the Khovanov homology groups $\text{Kh}(L)$ of the boundary links and the homotopy classes of chain maps between them. The *admissibility geometry* is

the coherence conditions ensuring that all extended movie moves commute up to chain homotopy: this is the constraint guaranteeing that the assignment of chain maps to movies is well-defined on isotopy classes.

5.2 Process primacy as a mathematical principle

The deepest shared feature of the three theories is that in each case the fundamental object is a *transformation class* rather than a *configuration*. A weighted one-foam is not primarily a graph with labels; it is a class of configurations related by admissible elementary cobordisms. A nilpotent pair (f, g) is not primarily a matrix pair; it is a point in the nilpotent cone distinguished from all other points by the accessibility structure of its balanced and unbalanced sectors. An immersed surface cobordism is not primarily a geometric shape; it is a movie equivalence class under the rewrite grammar of extended Carter–Saito moves.

In each case, the *object* emerges from the *transformation algebra* as a stable equivalence class, and the *invariants* — SAF class, nilpotency degree, Khovanov homology — are exactly what survives the identification under admissible moves. This is what we mean by the claim that constraint-first systems do not derive coherence from objects; rather, objects emerge as stable equivalence classes of admissible transformations.

5.3 Obstruction sensitivity as a recurring invariant principle

Each theory also shares what we term *obstruction sensitivity*: the invariants of the system encode not positive information about configurations but negative information about which transformations cannot be performed. The SAF class of a weighted foam records what cannot be undone by any sequence of elementary cobordisms. The nilpotency degree records how many round-trips are required to exhaust all accessible structure. The Khovanov homology records which linking and knotting data cannot be eliminated by any concordance.

This is characteristic of all genuine topological and homological invariants: they are not descriptive of positive structure but *obstructive* of positive transformation. A class in $H_1(\text{Aut}_{\text{IET}}, \mathbb{Z})$ obstructs the lifting of a transformation to a product of commutators. A nonzero element in $\text{Kh}(L)$ obstructs the existence of a null-cobordism of L in the ball. A nonzero nilpotency degree obstructs the existence of a nonzero vector surviving all iterated round-trips.

Obstruction sensitivity is the mathematical signature of constraint-first geometry: the invariants are defined by what the constraint system excludes rather than

by what it permits.

5.4 Formal introduction of CLIO

We now crystallize the operational concept that has been guiding the interpretations throughout this essay into a formal definition. The Constraint-Leveraged Inference Operator has appeared informally in each domain: as the compression map from foam configurations to SAF classes, as the nilpotency filter selecting admissible transport pairs, and as the chart projection encoding immersed surfaces in reduced dimension. Its formal role is that of a constrained projection preserving admissibility invariants under trajectory compression.

Definition 5.1 (CLIO — Constraint-Leveraged Inference Operator). Let $\mathcal{X}$ be a state or configuration space, $\mathcal{A}$ a system of local admissibility constraints (compatibility laws, cobordism moves, nilpotency conditions, movie-move equivalences), and $\mathcal{M}$ a compressed invariant space. A *Constraint-Leveraged Inference Operator* is a map

$$\text{CLIO}_{\mathcal{A}} : \mathcal{X} \rightarrow \mathcal{M}$$

satisfying the following conditions:

1. *Admissibility preservation.* $\text{CLIO}_{\mathcal{A}}$ is constant on $\mathcal{A}$ -equivalence classes: if $x \sim_{\mathcal{A}} x'$ then

$$\text{CLIO}_{\mathcal{A}}(x) = \text{CLIO}_{\mathcal{A}}(x').$$

2. *Obstruction faithfulness.* The fibers of $\text{CLIO}_{\mathcal{A}}$ are exactly the $\mathcal{A}$ -equivalence classes:

$$\text{CLIO}_{\mathcal{A}}(x) = \text{CLIO}_{\mathcal{A}}(x') \quad \text{if and only if} \quad x \sim_{\mathcal{A}} x'.$$

3. *Singular extension.* $\text{CLIO}_{\mathcal{A}}$ extends to a class of controlled singular inputs

$$\hat{\mathcal{X}} \supset \mathcal{X}$$

in which singularities are treated as structured morphism types carrying definite invariant data rather than as excluded configurations.

This definition is deliberately abstract. Its instances are the concrete maps studied in the three source papers: $\psi : \text{Cob}_{>0}^1 \rightarrow \mathbb{R} \wedge_{\mathbb{Q}} \mathbb{R}$ in foam theory (Theorem 2.1), the balanced-vector bijection of Theorem 3.2 in nilpotent transport theory, and the extended 2-functor $\tilde{\kappa}$ in immersed cobordism theory.

Each instance satisfies the three conditions of Definition 5.1. In foam theory: ψ is constant on cobordism classes by construction, is an isomorphism (hence obstruction-faithful), and the planar unoriented extension to $Z^2(\mathbb{R}_{>0})$ provides the

singular extension via the antisymmetric bracket. In nilpotent transport theory: the bijection of Theorem 3.2 identifies the nilpotent cone as the admissibility-preserved image, the balanced/unbalanced dichotomy is obstruction-faithful, and the Boolean semiring extension provides singular extension via DAG structure. In immersed cobordism theory: $\tilde{\kappa}$ is constant on isotopy classes by the movie-move theorem, is well-defined on homotopy classes of chain maps, and the singular extension condition is exactly the addition of node morphism types A and B .

5.5 The RSVP framework as admissibility meta-language

We close with a remark on the role of RSVP in this essay. The Relativistic Scalar-Vector Plenum framework has appeared throughout not as a primary mathematical object seeking validation from topology but as a *meta-interpretive language* for recognizing admissibility geometry across independent mathematical systems. RSVP provides the vocabulary for naming what the three source papers independently formalize: process- primary ontology, obstruction-sensitive invariants, and admissibility-governed coherence.

The RSVP framework’s central claim — that physical and semantic coherence arises from admissible trajectory equivalence rather than from intrinsic object properties — is precisely the mathematical content of Theorem 2.1, Theorem 3.2, and the main theorem of [3], each formulated in their own domain. The present essay argues that this convergence is not coincidental. The three papers are independently discovering the same constraint-first geometry because that geometry is a fundamental feature of any mathematical system in which coherence is preserved by transformation rather than configuration.

STRUCTURAL EQUIVALENCE PRINCIPLE 5.2 (Constraint-First Process Geometry). *Coherent structure in a constraint-first system is not a property of configurations but of transformation equivalence classes: objects emerge as stable equivalence classes of admissible local operations, invariants encode obstruction to simplification under admissible moves, and the admissibility geometry — the system of local compatibility laws — is the primary datum from which all other structure is derived. The cobordism group of weighted foams, the nilpotent cone of bidirectional transport pairs, and the extended Khovanov 2-functor on immersed cobordisms are three independent mathematical instantiations of this single architectural principle.*

6 Conclusion

We have traced a single architectural pattern through three domains of contemporary mathematics: the foam cobordism theory of Im and Khovanov, the nilpotent pair theory of Chen, Im, Khovanov, Lillja, and Rugo, and the immersed surface cobordism theory of Carter, Cooper, Khovanov, and Krushkal. In each domain, the fundamental mathematical entities are not configurations but equivalence classes of admissible local transformations. Invariants are defined by obstruction to simplification. Singularities carry structured algebraic information rather than representing failures of theory. Global coherence is guaranteed not by metric embedding but by the consistent closure of local compatibility laws.

The Constraint-Leveraged Inference Operator, introduced formally in Definition 5.1, abstracts this common architecture: it is any map from a configuration space to a compressed invariant space that is constant on admissibility equivalence classes, faithful to the obstruction structure of those classes, and extensible to controlled singular inputs carrying definite invariant data.

The RSVP framework provides the interpretive language for recognizing this architecture across domains. Its central claim — that physical and semantic coherence is an admissibility-geometric phenomenon rather than an intrinsic object property — is independently confirmed by each of the three source papers. The convergence is not imposed by external metaphysical framing but reflects a genuine structural feature of constraint-governed mathematical systems.

The deepest implication is philosophical as much as mathematical. A constraint-first ontology inverts the usual direction of explanation. It does not ask: “Given objects with intrinsic properties, which transformations are admissible?” It asks instead: “Given admissibility laws, which objects are stably distinguishable?” The answer — the orbits, invariants, and equivalence classes we have examined — constitutes not merely a different calculation but a fundamentally different conception of what mathematical structure is.

Constraint-first systems do not derive coherence from objects. Objects emerge as stable equivalence classes of admissible transformations.

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